

Weak and Strong T_{L^p} for Second Countable Locally Compact Groups

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Status

This is a likely valid strong partial result for arXiv:2310.18136. The source paper records that weak property T_{L^p} and property T_{L^p} are known to coincide for countable discrete groups, and says that the corresponding locally compact question is open. The result below proves the second countable locally compact case. Arbitrary non-second-countable locally compact groups are not claimed.

Source Question

The source is E. M. Elkiær and S. Pooya, *Property (T) for Banach algebras*, arXiv:2310.18136 [4]. In the remark following Theorem 4.3, after discussing $F_{\lambda_p}^*(G)$, it states that for countable discrete groups the equivalence between weak property T_{L^p} and property T_{L^p} is known, but that it is open whether this is true for general locally compact groups.

Theorem

Theorem 1. *Let G be a second countable locally compact group and let $1 \leq p < \infty$. Then G has weak property T_{L^p} if and only if it has property T_{L^p} .*

Here L^p denotes the class of complex L^p -spaces over sigma-finite measure spaces, and representations are strongly continuous isometric representations.

Proof Intuition

The discrete proof of Elkiær [3] uses circle-valued measurable cocycles. If coboundaries were closed for every ergodic p.m.p. action, the open mapping theorem for Polish groups would force strong ergodicity. For a group without property T , Connes-Weiss gives an ergodic action that is not strongly ergodic, hence a cocycle in the closure of the coboundaries but not itself a coboundary. This cocycle twists the Koopman representation on L^p ; approximating coboundaries give almost invariant unit vectors, while non-coboundary prevents invariant vectors.

For locally compact second countable G , the same mechanism works after replacing the countable product topology on cocycles by compact-open convergence in measure. Compact sets play exactly the role of finite subsets in the definition of almost invariant vectors.

Cocycles

Let $G \curvearrowright (\Omega, \mu)$ be an ergodic probability-measure preserving action on a standard probability space. We write

$$(gf)(\omega) = f(g^{-1}\omega)$$

for the induced action on measurable functions. Let $L^0(\Omega; \mathbb{T})$ be equipped with convergence in measure; equivalently, for circle-valued functions, with any L^p -metric.

A $\mathbb{T}$ -valued 1-cocycle is a continuous map

$$c : G \rightarrow L^0(\Omega; \mathbb{T}), \quad g \mapsto c_g,$$

where continuity is for convergence in measure, satisfying

$$c_{gh}(\omega) = c_g(\omega)c_h(g^{-1}\omega)$$

for all $g, h \in G$, almost everywhere. Let $Z^1(G \curvearrowright \Omega; \mathbb{T})$ be the group of these cocycles with the compact-open topology:

$$c_i \rightarrow c \iff \sup_{g \in K} \|c_{i,g} - c_g\|_{L^p(\mu)} \rightarrow 0 \quad \text{for every compact } K \subset G.$$

Because G is second countable locally compact and $L^0(\Omega; \mathbb{T})$ is Polish, this compact-open topology makes Z^1 a Polish group; the cocycle identity defines a closed subgroup of the Polish group $C(G, L^0(\Omega; \mathbb{T}))$.

For $\phi \in L^0(\Omega; \mathbb{T})$, define

$$b_\phi(g, \omega) = \phi(\omega) \overline{\phi(g^{-1}\omega)}.$$

The set B^1 of such coboundaries is a subgroup of Z^1 . The map $\beta : L^0(\Omega; \mathbb{T}) \rightarrow B^1$, $\beta(\phi) = b_\phi$, is a continuous surjective homomorphism. Continuity into the compact-open topology follows from strong continuity of the action on $L^p(\Omega)$, uniformly on compact subsets, and from the fact that all functions involved are $\mathbb{T}$ -valued. Since the action is ergodic, the kernel consists precisely of the constants.

Lemma 1. *If $B^1(G \curvearrowright \Omega; \mathbb{T})$ is closed in $Z^1(G \curvearrowright \Omega; \mathbb{T})$, then the action is strongly ergodic.*

Proof. If B^1 is closed, then it is Polish. The open mapping theorem for Polish groups applied to $\beta : L^0(\Omega; \mathbb{T}) \rightarrow B^1$ shows that β is open. Fix $0 < \varepsilon < 1/2$ and set

$$U_\varepsilon = \{\phi \in L^0(\Omega; \mathbb{T}) : \inf_{z \in \mathbb{T}} \|\phi - z\|_{L^p(\mu)} < \varepsilon\}.$$

Then $\beta(U_\varepsilon)$ is an open neighborhood of the identity in B^1 . Hence there are a compact set $K \subset G$ and $\delta > 0$ such that, whenever $\phi \in L^0(\Omega; \mathbb{T})$ satisfies

$$\sup_{g \in K} \|b_\phi(g) - 1\|_{L^p(\mu)} < \delta,$$

we have $b_\phi \in \beta(U_\varepsilon)$. This means that after multiplying ϕ by a constant in $\mathbb{T}$, it is within ε of 1 in L^p .

Suppose the action were not strongly ergodic. Then there is a net of measurable sets A_i with $\mu(A_i) = 1/2$ such that

$$\sup_{g \in K} \mu(gA_i \triangle A_i) \rightarrow 0$$

for every compact $K \subset G$. Put $\phi_i = 1_{A_i} - 1_{\Omega \setminus A_i}$. Then

$$\|b_{\phi_i}(g) - 1\|_{L^p}^p = 2^p \mu(gA_i \triangle A_i),$$

so $b_{\phi_i} \rightarrow 1$ in B^1 . By the previous paragraph, ϕ_i is eventually within ε of a constant circle-valued function. This is impossible for $\varepsilon < 1$, since every ϕ_i takes the two values 1 and -1 on sets of measure $1/2$. Thus the action is strongly ergodic. $\square$

Lemma 2. *Let $c \in Z^1(G \curvearrowright \Omega; \mathbb{T})$. Define*

$$(\pi_c(g)f)(\omega) = c_g(\omega)f(g^{-1}\omega), \quad f \in L^p(\Omega).$$

Then π_c is a strongly continuous isometric representation of G on $L^p(\Omega)$.

Moreover:

1. *if $c \in \overline{B^1}$, then π_c has almost invariant unit vectors;*
2. *if $c \notin B^1$, then π_c has no non-zero invariant vector.*

Proof. The cocycle identity gives the representation identity, and the values in $\mathbb{T}$ make the operators isometric. Strong continuity follows from continuity of $g \mapsto c_g$ in measure and continuity of the Koopman representation; bounded convergence upgrades this to L^p -continuity on circle-valued multipliers.

Assume $c \in \overline{B^1}$, and choose a net $\phi_i \in L^0(\Omega; \mathbb{T})$ with $b_{\phi_i} \rightarrow c$ in compact-open topology. Each ϕ_i is a unit vector in $L^p(\Omega)$, and for every compact $K \subset G$,

$$\begin{aligned} \sup_{g \in K} \|\pi_c(g)\phi_i - \phi_i\|_{L^p} &= \sup_{g \in K} \|c_g(\omega)\phi_i(g^{-1}\omega) - \phi_i(\omega)\|_{L^p_\omega} \\ &= \sup_{g \in K} \|c_g - b_{\phi_i}(g)\|_{L^p} \rightarrow 0. \end{aligned}$$

Thus π_c has almost invariant unit vectors.

Conversely, suppose $0 \neq f \in L^p(\Omega)$ is invariant. Then $|f|$ is invariant under the original p.m.p. action. Ergodicity implies that $|f|$ is almost everywhere constant and non-zero. Hence $\phi = f/|f|$ is circle-valued and

$$c_g(\omega)\phi(g^{-1}\omega) = \phi(\omega)$$

for every g , almost everywhere. Equivalently $c = b_\phi$, contrary to $c \notin B^1$. Therefore no non-zero invariant vector exists when c is not a coboundary. $\square$

Proof of Theorem 1. Property T_{L^p} plainly implies weak property T_{L^p} . For the converse, assume that G does not have property T_{L^p} . By the Bader-Furman-Gelander-Monod theorem for second countable locally compact groups [1], this means that G does not have Kazhdan's property T .

By the locally compact Connes-Weiss-Schmidt strong-ergodicity characterization of property T , there is an ergodic p.m.p. action of G on a standard probability space which is not strongly ergodic. By Lemma 1, B^1 is not closed in Z^1 . Choose

$$c \in \overline{B^1} \setminus B^1.$$

Lemma 2 gives a strongly continuous isometric representation π_c on the L^p -space $L^p(\Omega)$ with almost invariant unit vectors and no non-zero invariant vector. Hence G fails weak property T_{L^p} . This proves that weak property T_{L^p} implies property T_{L^p} . $\square$

Consequence for the Source Paper

The source paper proves that for discrete groups $F_{\lambda_p}^*(\Gamma)$ detects T_{L^p} . It also remarks that, for non-discrete IN groups and $p \neq 2$, the same method gives only weak T_{L^p} for the group. Theorem 1 upgrades that weak conclusion to T_{L^p} whenever the IN group is second countable.

Limitations and Novelty Check

This packet does not treat arbitrary non-second-countable locally compact groups. The proof uses second countability to make the compact-open cocycle space Polish and to invoke the usual standard probability-space form of the strong-ergodicity characterization.

Cheap run indexes had no prior packet for arXiv:2310.18136. The source paper cites arXiv:2403.05312 [3] for countable discrete groups and explicitly records the locally compact case as open. Bounded web searches on 2026-06-30 for exact phrases including

“weak property” “ T_{L^p} ” “locally compact”

and

“ $F_{\lambda_p}^*(G)$ ” “property” “T” “non-discrete”

found no later settlement.

Human Review Recommendation

Review as a likely valid strong partial result. The main points to check are the compact-open Polish topology on Z^1 , the uniform-on-compact continuity of the coboundary map, and the exact locally compact form of the Connes-Weiss-Schmidt theorem used in the final proof.

References

- [1] U. Bader, A. Furman, T. Gelander, and N. Monod, *Property (T) and rigidity for actions on Banach spaces*, Acta Math. 198 (2007), 57–105.
- [2] A. Connes and B. Weiss, *Property T and asymptotically invariant sequences*, Israel J. Math. 37 (1980), 209–210.
- [3] E. M. Elkiær, *Weak property T_{L^p} for discrete groups*, arXiv:2403.05312.
- [4] E. M. Elkiær and S. Pooya, *Property (T) for Banach algebras*, arXiv:2310.18136.
- [5] K. Schmidt, *Asymptotically invariant sequences and an action of $SL(2, \mathbb{Z})$ on the 2-sphere*, Israel J. Math. 37 (1980), 193–208.